Machine Learning for Social Data Science

Problem Set 2

How effective is BERTopic at classifying Community Ownership Fund Projects? (1,500 words)

Introduction

The UK is one of the most regionally unequal countries in the OECD (McCann, 2016, 2020), with significant disparities between London and the South East and much of the rest of the country in productivity, infrastructure, and transport (Davenport and Zaranko, 2020; Fai and Tomlinson, 2023). These inequalities are reflected in social outcomes: life expectancy in Blackpool and Middlesbrough is nearly a decade lower than in parts of Surrey (ONS, 2024). Such gaps reflect 'left-behind' communities shaped by deindustrialisation, austerity, and perceived political neglect (Gray and Barford, 2018; Hooper, 2023), contributing to the declining trust in the state's capacity to deliver outside traditional centres of power. Against this backdrop, Levelling Up emerged as both a corrective policy intervention and a visible sign of renewed state attention to 'left-behind' communities.

This essay builds on my undergraduate dissertation, which explored the impact of Levelling Up funding on turnout and Conservative vote share in the 2024 General Election through a clientelist framework. It found that constituencies lost or newly won by the Conservatives in 2019 were more likely to receive funding than non-Conservative or held seats (*Appendix A*). On average, each £10 million allocated was associated with a 0.71% fall in turnout and 0.5% decline in Conservative vote share. However, the effects of different types of projects remain unclear. Evidence from the US suggests that every 1% increase in federal funding increases incumbents' vote share by 1.8% in closely contested districts (Kriner and Reeves, 2012, p. 359), particularly when directed towards infrastructure and public services (Hersh, 2015).

To address this gap, this essay applies a topic model to structured text data to generate project labels. The structure of this essay is as follows: a data and methods section, followed by findings, and finally, a conclusion.

Data and Methods

To generate these labels, topic modelling is used, an unsupervised machine learning method that identifies latent themes in text. This essay uses BERTopic, which improves on traditional 'bag-of-words' approaches by combining embeddings with clustering (Grootendorst, 2022). First, documents are cleaned and transformed into embeddings to capture semantic meaning. UMAP then reduces dimensionality while preserving global and local structure, outperforming T-SNE and PCA in maintaining meaningful relationships (Gong, 2025). HDBSCAN clusters documents into topics, allowing flexible, noise-aware groupings without requiring a pre-specified number of clusters, unlike K-Means.

The text is then vectorised, converting text into numerical representations based on word frequencies, and then applies TF-IDF to identify words specific to each topic by weighting terms according to their importance within clusters. Representation modelling refines topic labels by using the embedding model to select keywords that are both distinctive to the topic and semantically coherent. Finally, BERTopic integrates these steps, producing coherent, interpretable topics that generally outperform other unsupervised topic modelling techniques (Koráb, 2025).

Due to limited existing research, there is no comprehensive public dataset on Levelling Up projects; therefore, this essay uses data collected from my undergraduate dissertation. Projects were initially assigned labels based on their stated purpose during the original data collection; however, the reliability of these labels is limited: categories evolved over time, and earlier entries were not systematically revised to reflect newly introduced labels, leading to potential inconsistencies.

The text corpus comprises individual project documents compiled from a variety of sources, including local media reports, government publications, and social media posts. The documents establish the project's purpose, location, and completion status. There are 338 projects from the Community Ownership Fund (COF); other funding streams, such as the Levelling Up Fund, Long Term Plan for Towns, and the Towns Fund, have been excluded due to time constraints and collection challenges.

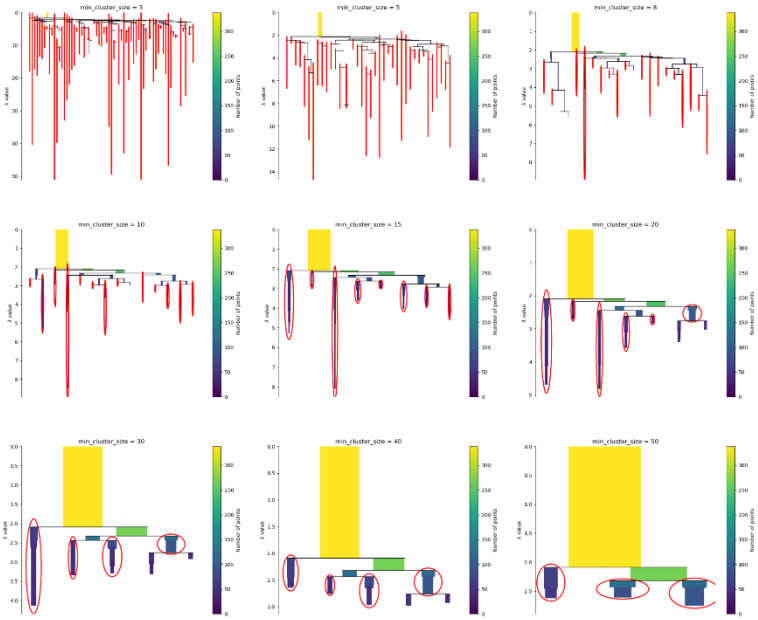
This essay addresses three research question:

1. *Can BERTopic generate coherent and distinct topics for COF projects?*
2. *How accurately does BERTopic assign labels to COF projects compared to manual classification?*
3. *Do the topics generated by BERTopic provide substantively meaningful insights into the nature and purpose of COF projects?*

The unit of analysis and observation is individual projects. Internal validity is strengthened by using systematic text cleaning, embeddings, UMAP for dimensionality reduction, and HDBSCAN clustering, which come together to minimise noise and improve the consistency and coherence of the topic labels. External validity is limited, as the results are specific to COF projects and may not be generalisable to other funding streams or redistributive policies. Ethical concerns are minimal: all data collected is publicly available, and the risk of harm to projects or communities is negligible. To ensure research integrity, a GitHub repository link is provided (*Appendix B*).

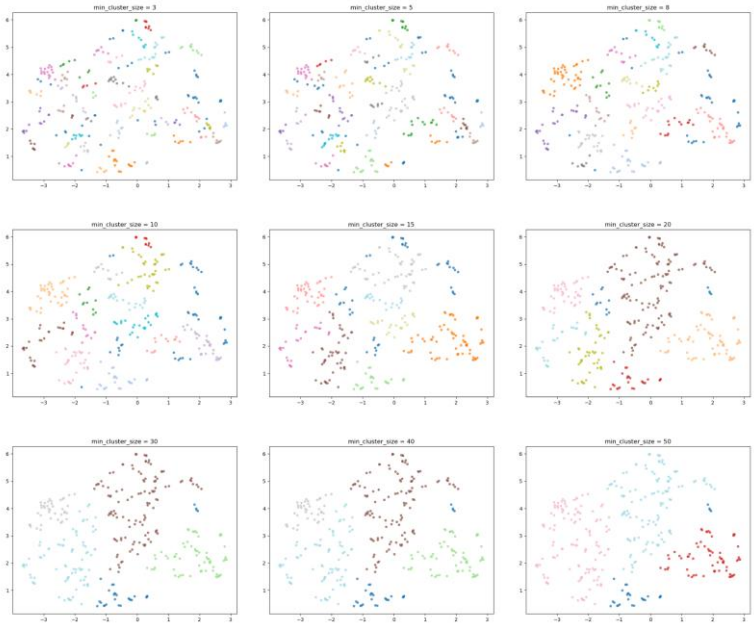
Sentence embeddings were generated using the 'all-mini-L6-v2' model, chosen for its efficiency and strong performance in capturing semantic meaning across short project descriptions. Alternative embedding models were also tested to evaluate robustness. UMAP reduced dimensionality with 5 neighbours and a minimum distance of 0.00, preserving both local and global structure. HDBSCAN clustering parameters were selected using condensed trees and scatterplots: Figures 1 and 2 show the effect of varying minimum cluster size, while Figures 3 and 4 illustrate changes in minimum sample size.

Figure 1: Condensed Tree for Minimum Cluster Sizes



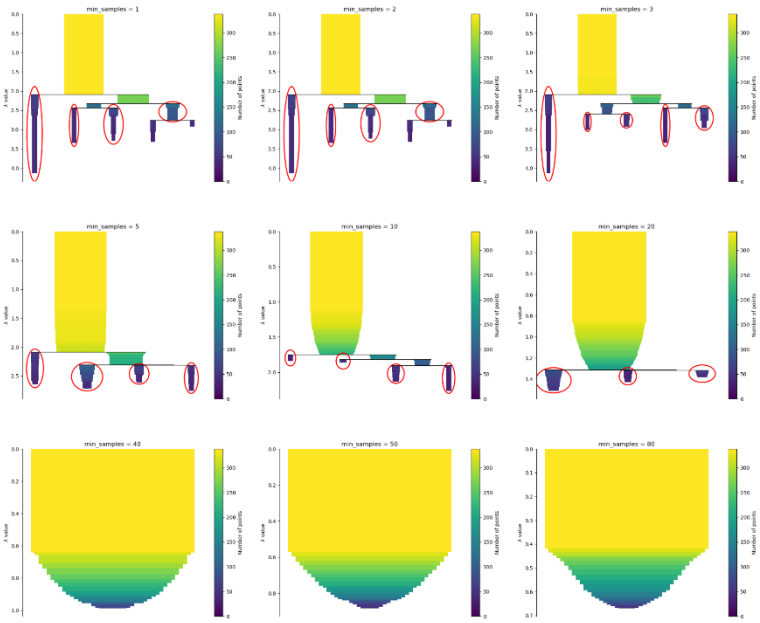
Note: Assuming minimum sample size of 1, euclidean metrics, and excess of masses algorithm

Figure 2: Scatter Plot for Minimum Cluster Sizes



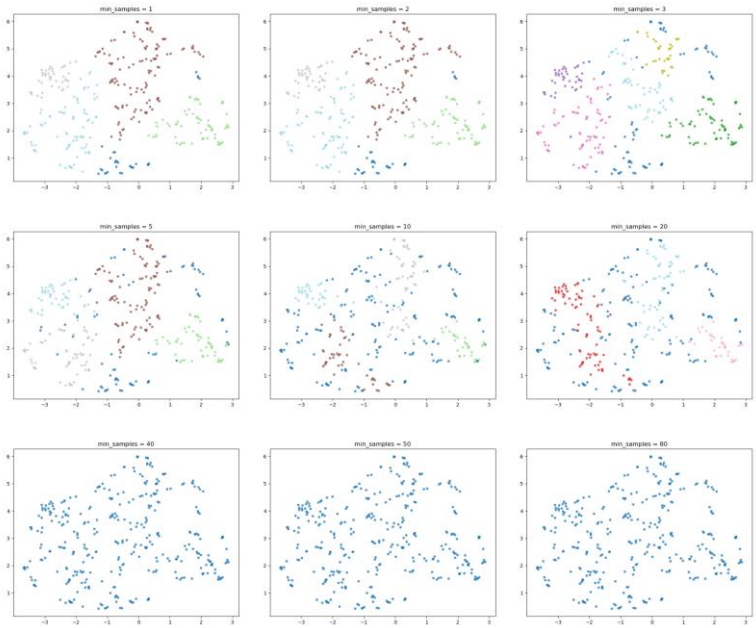
Note: Assuming minimum sample size of 1, euclidean metrics, and excess of masses algorithm

Figure 3: Condensed Tree for Minimum Sample Sizes



Note: Assuming minimum cluster size of 15, euclidean metrics, and excess of masses algorithm

Figure 4: Scatter Plot for Minimum Sample Sizes



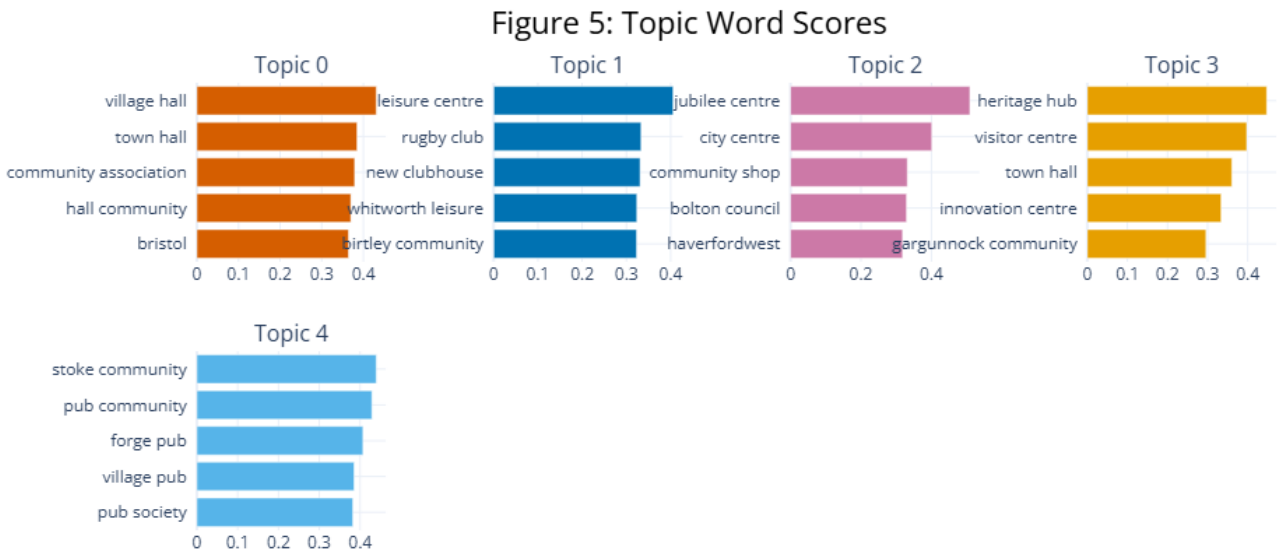
Note: Assuming minimum cluster size of 15, euclidean metrics, and excess of masses algorithm

While the plots recommend a minimum cluster size of 15 and a minimum sample size of 1 due to their ability to generate wide and stable trees, values of 30 and 3 were selected to produce more distinctive and semantically coherent topics. This decision is supported by a noise ratio of 0.1952, indicating a moderate number of unassigned points that improves reliability. The text vectoriser captures both uni- and bi-grams, while ignoring extremely rare or overly common terms, ensuring focus on informative project content.

Findings

The findings section analyses the model in three layers: first, an evaluation of the topics generated by the model, focusing on coherence, diversity, and the relationship between topics; second, an evaluation of the label assignment through F1, precision, recall, and accuracy metrics; and finally, a substantive evaluation of what sort of topics have been detected by the model using word clouds and what this tells us about COF projects.

Figure 5 shows high topic coherence overall, with tightly grounded and semantically consistent keywords within most clusters. Topics 0 (community spaces) and 1 (leisure and fitness) exhibit strong coherence through repeated references to halls, centres, and leisure. Topic 3 (heritage, tourism, and culture) is similarly well-defined, with clear cultural and visitor-oriented terms. Topic 4 (community pubs) is the most coherent, with highly specific and homogenous keywords. In contrast, topic 2 (Economic) is less cohesive, combining retail, administrative, and civic terms, indicating weaker topic consistency and greater thematic spread.



The heatmap below (Figure 6) indicates moderate topic diversity within a relatively high similarity range (0.80-1.0). The strongest overlap is between economic and community spaces (~0.95),

suggesting shared place-based language. Community pubs also show relatively high similarity with economic (~0.91), heritage, tourism, and culture, and community spaces (~0.90). Contrastingly, leisure and fitness is the most distinctive topic, with lower similarity scores with heritage, tourism, and culture (~0.80), and community pubs (~0.83). Overall, while topics are distinguishable, consistently high similarity scores indicate partial thematic convergence.

Figure 6: Topic Similarity Matrix

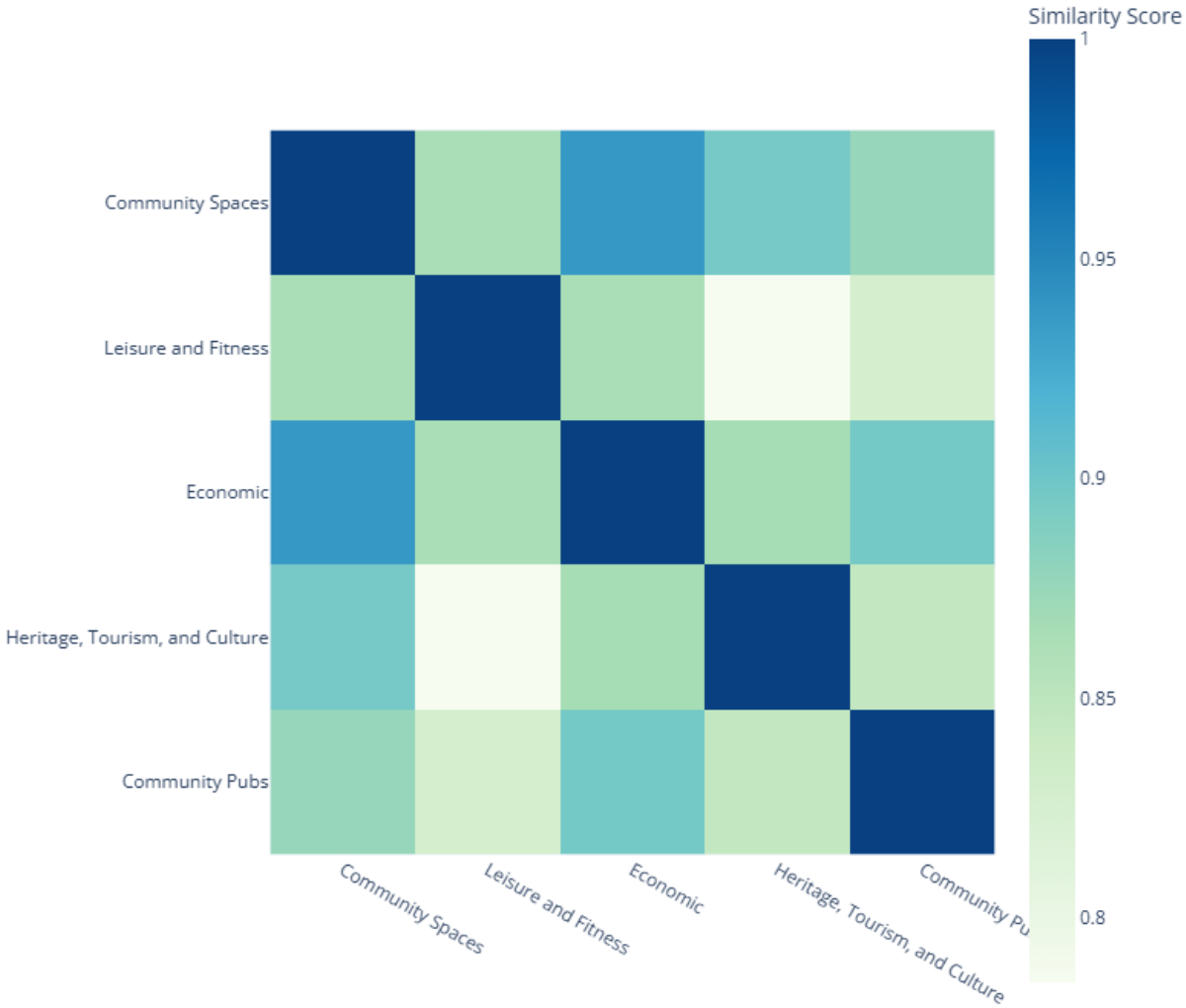
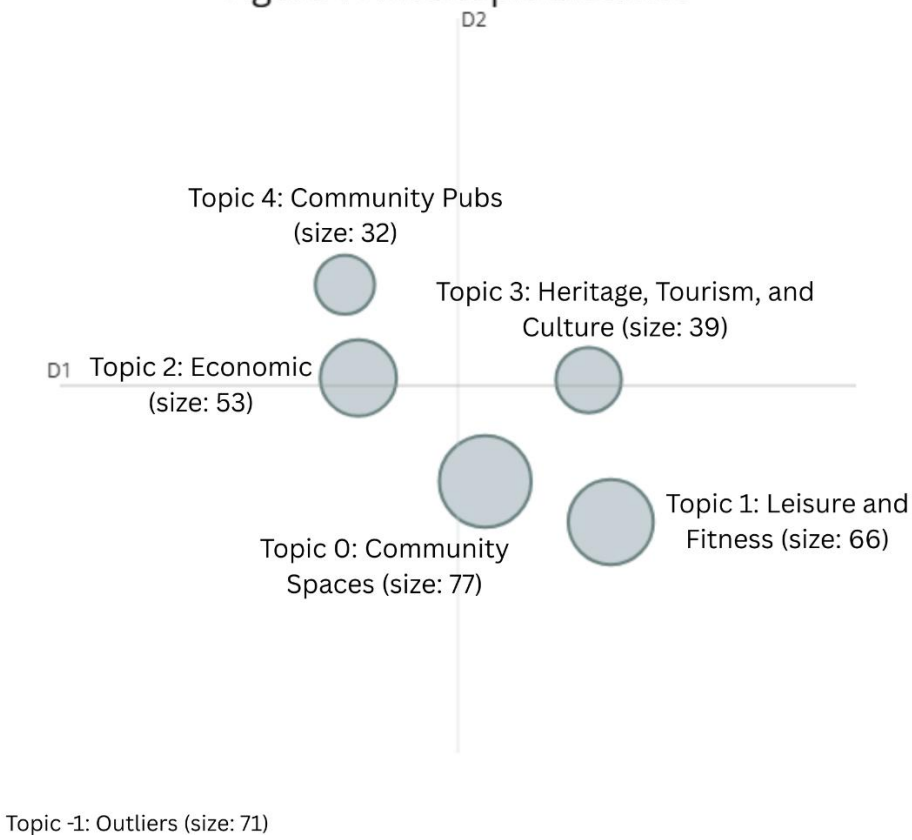


Figure 7 below shows moderate topic separation, with clusters occupying separate but close regions of semantic space. Topics 0 and 1 are positioned relatively close, suggesting shared language around local amenities. Topics 2 and 4 lie near each other, indicating an overlap between community assets and economic terms. On the other hand, topic 3 is more isolated, reflecting greater semantic distinctiveness. In total, 267 projects are assigned to topics, with 71 outliers, reaffirming reasonable clustering with some residual noise. Overall, distances reveal meaningful differentiation, though proximity between several topic points remains low.

Figure 7: Intertopic Distance



Next, a random sample of five projects per topic was manually inspected to assess whether projects were assigned appropriate labels with metrics reported in Figure 8. The model achieved an accuracy of 0.533, indicating that just over half of the assignments are correct, suggesting room for improvement. Precision (0.533) exceeds recall (0.474), meaning the model is better at correctly identifying relevant labels than capturing all relevant cases. The F1 score is 0.495, reflecting balanced but limited performance across topics. Figure 9 further shows that misclassified labels are concentrated in topic -1 (outliers), followed by topics 3 (heritage, culture, and tourism) and 0 (community spaces), indicating overlap between these themes and contributing to reduced recall.

Figure 8: Performance Table

	Score
F1 (Macro)	0.495
Precision (Macro)	0.533
Recall (Macro)	0.474
Accuracy	0.533

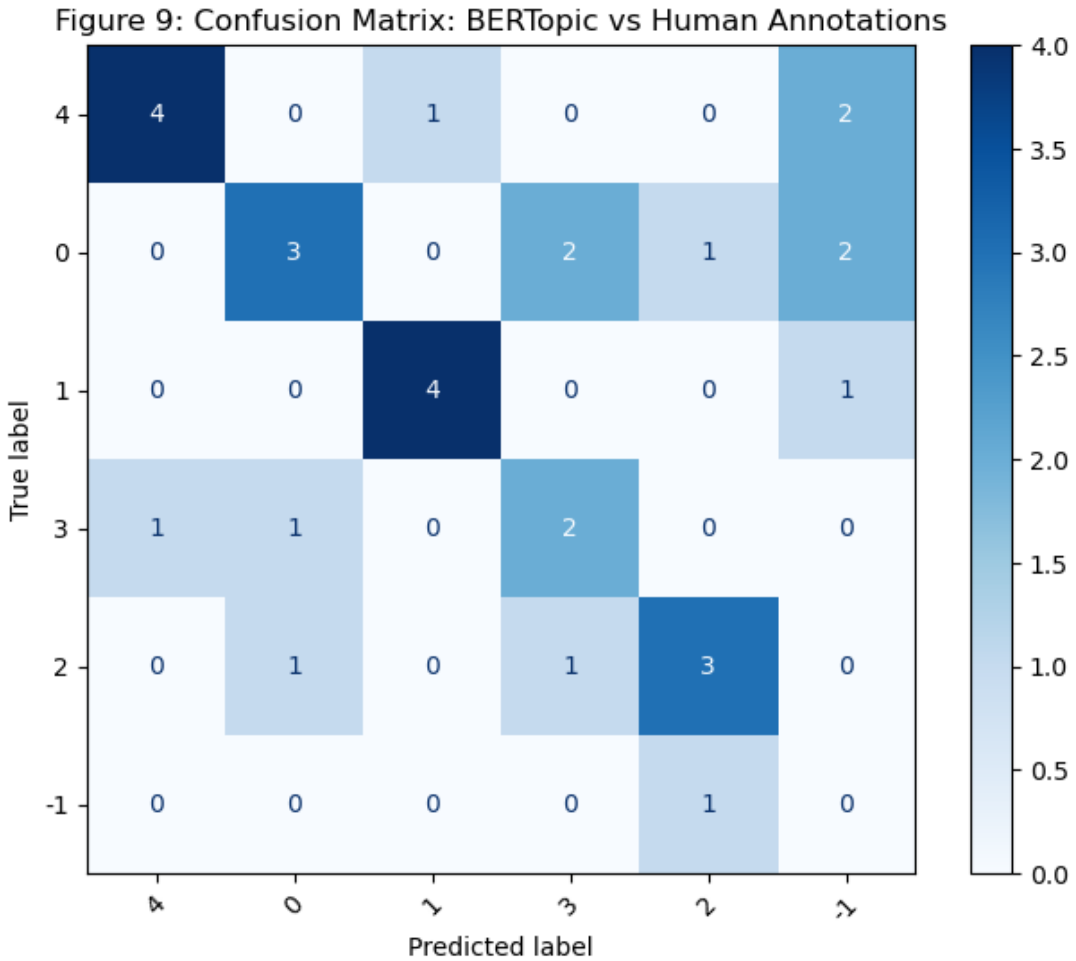


Figure 10 displays the topics' word clouds, which suggest that COF projects are strongly centred on community-owned infrastructure, such as community shops, sports clubhouses, and pubs. The emphasis on funding local venues, 'support services', and community indicates a focus on sustaining everyday amenities, preserving social hubs, and reinforcing community cohesion, rather than large-scale economic transformation or productivity growth.

Figure 10: Topic Wordclouds

Conclusion

Overall, the model produces reasonably coherent and interpretable topics, though high similarity between clusters indicates partial thematic overlap. Label evaluation shows moderate performance, with accuracy and F1 scores suggesting the model captures broad patterns but struggles with precise classification, particularly between closely related topics. Substantively, the topics reveal that COF projects are concentrated on maintaining community assets and local services rather than driving structural economic change. Future methodological research could refine preprocessing by removing location-specific language to improve topic coherence, and explore zero-shot models to assess whether they can generate labels from initial human classifications (Grootendorst, 2025). Empirically, expanding the corpus to include other funding streams would strengthen external validity and enable comparative analysis across funding streams. In conclusion, BERTopic demonstrates clear potential for generating meaningful labels, but its effectiveness is constrained by overlap between themes and limits in classification accuracy.

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Appendices

Appendix A – Findings

Figure A.1: Hierarchial Clustering

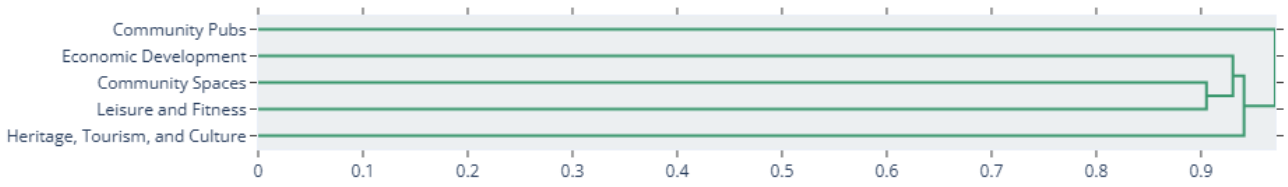
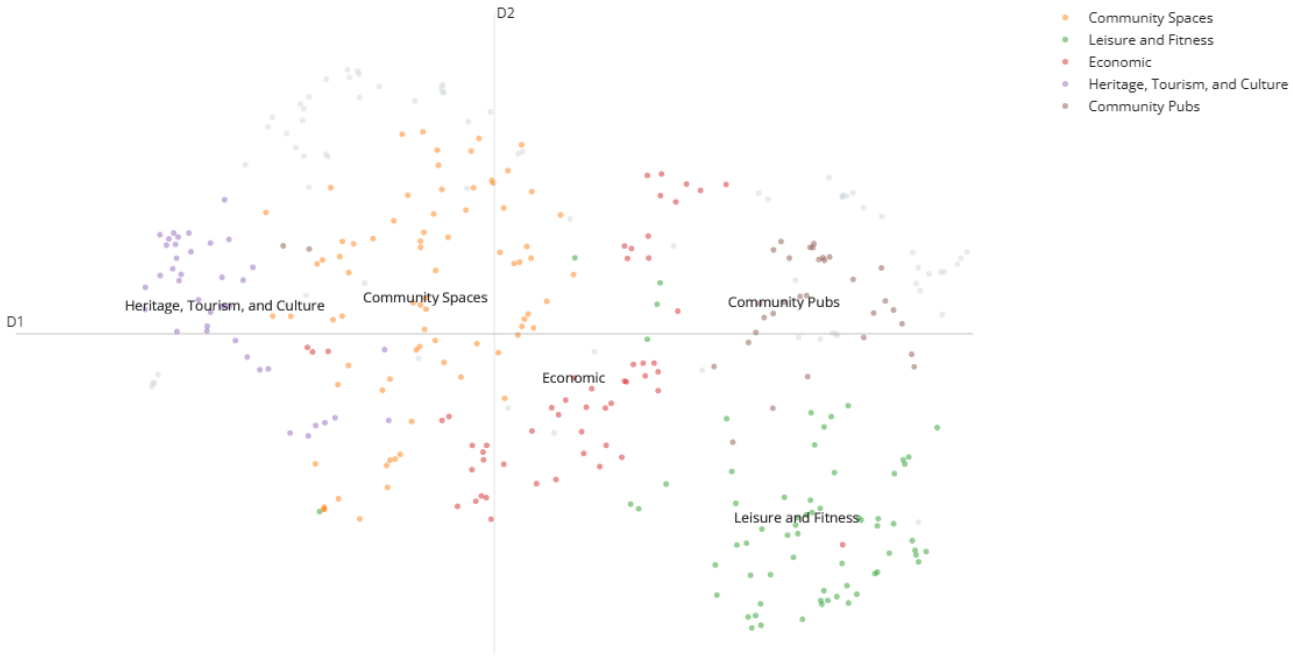


Figure A.2: Documents and Topics



Appendix B – GitHub Repository Link

https://github.com/RyanWhitehead-04/MLSDS-Problem-Set-2.git

Appendix C – Google Antigravity AI Agent Chat History

Chat Conversation

Note: _*This is purely the output of the chat conversation and does not contain any raw data, codebase snippets, etc. used to generate the output.*_

User Input

I have a dataset loaded in called ‘data’. This dataset contains textual data which includes URLs (starting with https://) and access dates (formatted as [Data Accessed: Date, month, year], for example [Date Accessed: 22nd August 2025]). The data is currently in the format of a list of dictionaries. The aim is to clean the ‘Text’ column by removing the URLs and the access dates.

Please help me:

1. Create a regex pattern to identify and remove any URLs starting with https://
2. Create a regex pattern to identify and remove any access dates in the format of [Date Accessed: date, month, year].
3. Write a loop to clean the raw text by removing the URLs and access dates from each entry
4. Add comments explaining each step of the process

Provide the code so I can copy and paste it into my notebook

Planner Response

Here is the complete Python code implementing the regex patterns and the loop to clean your text data.

I have used the built-in `re` (regular expression) module in Python, which is perfect for identifying and removing specific text patterns.

```python
import re

# 1. Create a regex pattern to identify URLs starting with https://
# 'https://' matches the literal string.
# '\S+' matches one or more non-whitespace characters (the rest of the URL).
url_pattern = r'https://\S+'
```

```
# 2. Create a regex pattern to identify access dates
# '\[' and '\]' are used to match the literal square bracket characters.
# 'Date Accessed:' matches the literal text.
# '.*?' non-greedily matches any characters (the date itself) up until the closing
bracket.
date_pattern = r'\[Date Accessed:.*?\]'

# 3. Write a loop to clean the raw text for each entry in the dataset
for entry in data:
    # Check if 'Text' exists in the dictionary and is a string type just to be safe
    if 'Text' in entry and isinstance(entry['Text'], str):

        # Extract the current raw text
        current_text = entry['Text']

        # Use re.sub() to substitute the URL pattern with an empty string ('')
        text_without_urls = re.sub(url_pattern, '', current_text)

        # Use re.sub() to substitute the Date pattern with an empty string ('')
        text_without_dates = re.sub(date_pattern, '', text_without_urls)

        # Strip removes any extra whitespace left at the start or end of the string
        cleaned_text = text_without_dates.strip()

        # 4. Update the dictionary with the fully cleaned text
        entry['Text'] = cleaned_text

# Optional: Print the first few entries to verify it worked
# print(data[:5])
```

Explanation of the Regex step-by-step:

1. **URLs (`r'https://\S+'`)**: The `\S+` part is crucial here. `\S` means "any character that is **not** a space" and `+` means "one or more times". This perfectly captures a URL until it hits a space.
2. **Dates (`r'\[Date Accessed:.*?\]'`)**: We have to put backslashes `\` in front of the square brackets `\[` `\]` because square brackets usually have a special meaning in regex. The `.*?` means "match absolutely anything here, but stop as soon as you hit the literal `]`".

You can copy and paste this directly into a Jupyter Notebook cell right underneath where your `data` variable is loaded! Let me know if you run into any issues.